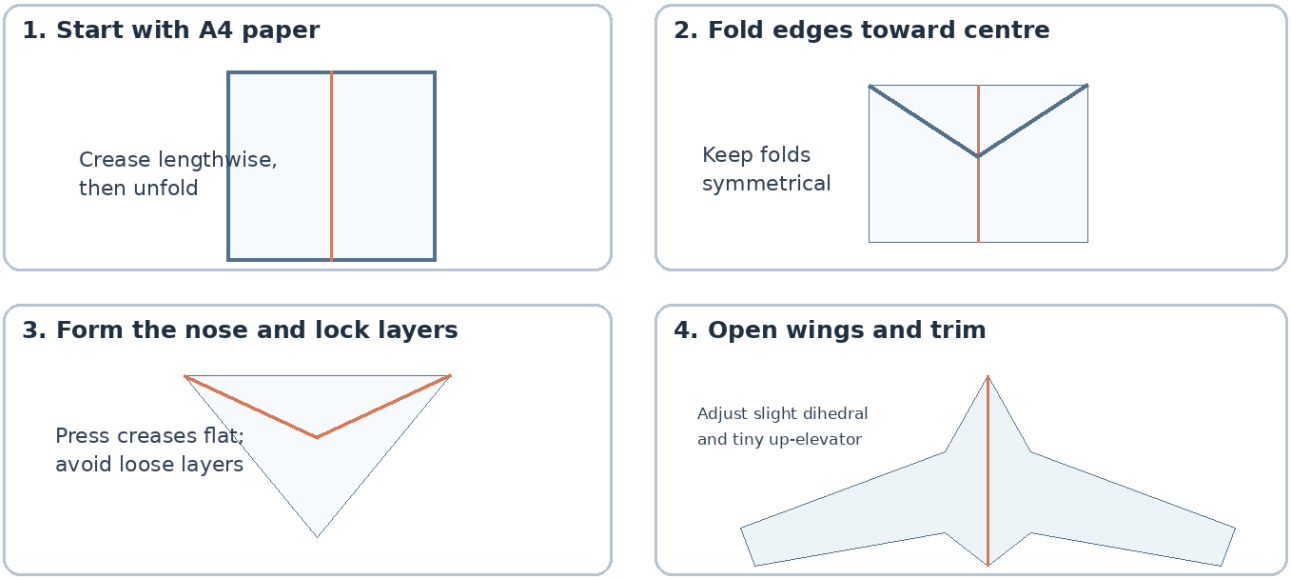
Experimental Study of the Suzanne Paper Glider

Design reference: John Collins, WIRED — “Aerodynamics Explained by a World Record Paper Airplane Designer”

Objective. Evaluate the flight duration of the Suzanne paper glider using five recorded stopwatch trials. This report focuses on aerodynamic principles and measured results.

Suzanne design schematic



The illustration is a conceptual schematic, not a photograph of the student’s actual plane. For exact folding details, consult the cited tutorial.

Aerodynamic principles

Forces in flight. Lift acts approximately perpendicular to the airflow, drag opposes motion, and weight acts downward. The hand launch supplies the initial energy; the glider has no sustained propulsion. During the glide, it loses altitude as gravitational potential energy is converted into motion and overcomes drag.

Wing area and wing loading. Wing loading is aircraft weight divided by lifting area. For the same paper mass, greater effective wing area can support slower flight, but does not by itself guarantee longer airtime; drag, trim, launch speed and stability also affect performance.

Dihedral and pitch trim. Slightly raised wings can help the model recover from a roll. A small upward bend at the rear edges changes pitch trim. Excessive bend can increase drag or cause a stall, while insufficient trim can lead to a dive.

Centre of gravity and glide ratio. A forward centre of gravity generally supports steady tracking, while wing forces and trim establish pitching balance. Glide ratio is horizontal distance divided by altitude lost. This experiment measured time rather than distance, so glide ratio was not calculated.

Reynolds number. Reynolds number compares inertial and viscous effects in airflow and depends on speed and characteristic length. A paper glider operates at a relatively low Reynolds number, so its behaviour should not be inferred directly from full-scale aircraft data.

Flight-time experiment

Five stopwatch readings supplied by the student were transcribed as follows. Launch height, launch speed, and environmental conditions were not specified in the supplied evidence.

Trial	Duration (s)
1	3.55
2	5.75
3	2.95
4	3.20
5	2.73

Mean	3.64

Mean flight duration: (3.55 + 5.75 + 2.95 + 3.20 + 2.73) / 5 = 18.18 / 5 = **3.636 s** ≈ **3.64 s**.

Conclusion

Across five recorded trials, the Suzanne glider had a mean observed flight time of **3.64 seconds**. The measured durations ranged from 2.73 s to 5.75 s. Since the available evidence does not document all launch and environmental conditions, this result should be treated as a preliminary measurement.

References

[1] WIRED, “Aerodynamics Explained by a World Record Paper Airplane Designer | Level Up,” John Collins, 13 Oct 2020. https://youtu.be/3KqjRPV9_PY.

[2] Experimental data: five stopwatch screenshots supplied by the student; durations transcribed as 3.55, 5.75, 2.95, 3.20 and 2.73 seconds.